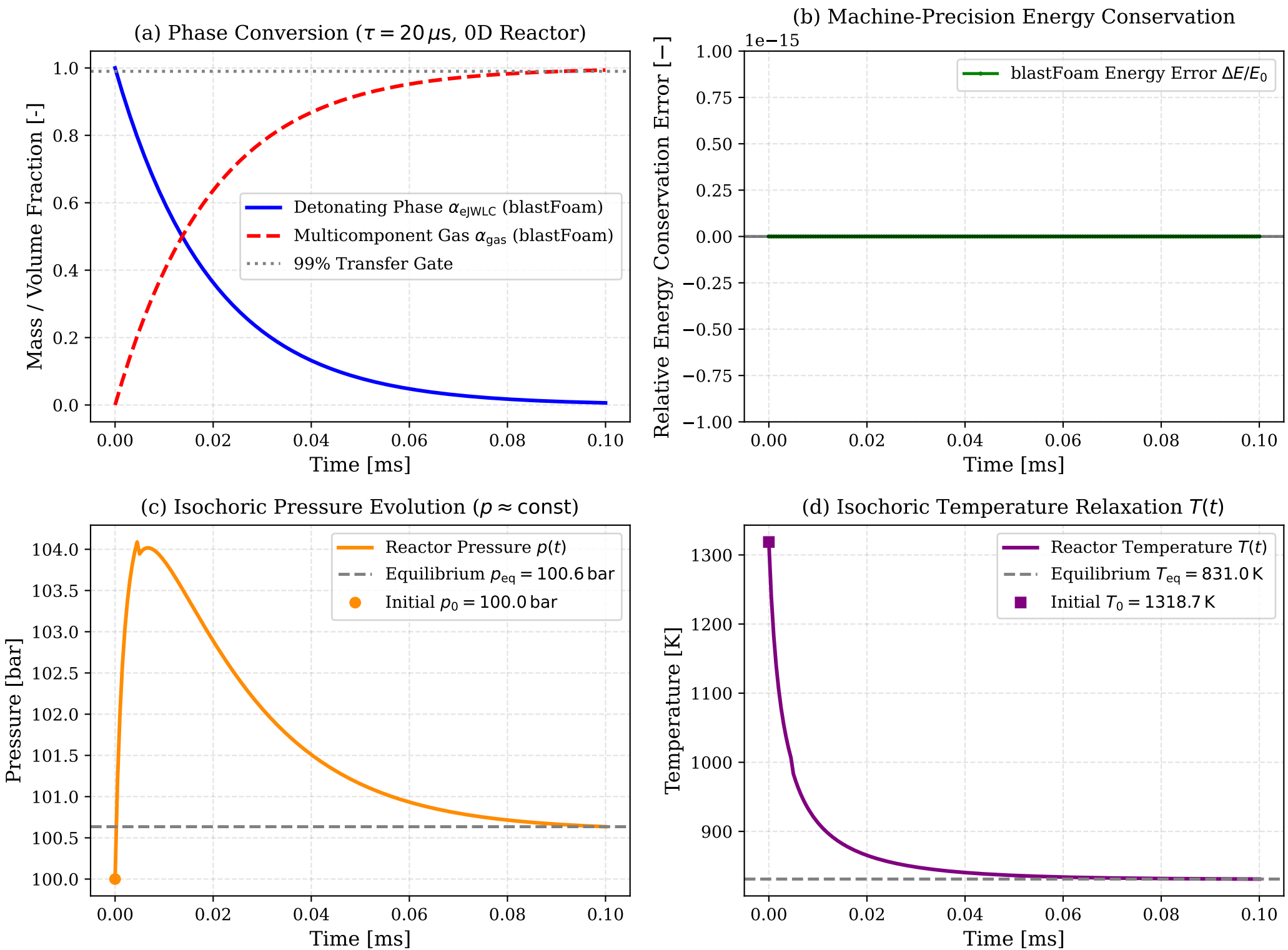
(a) Phase Conversion ($\tau = 20\,\mu$s, 0D Reactor)
Mass / Volume Fraction [-]
Time [ms]
Detonating Phase $\alpha_{\mathrm{eJWLC}}$ (blastFoam)
Multicomponent Gas $\alpha_{\mathrm{gas}}$ (blastFoam)
99% Transfer Gate
(b) Machine-Precision Energy Conservation
Relative Energy Conservation Error [−]
1e−15
blastFoam Energy Error $\Delta E/E_0$
Time [ms]
(c) Isochoric Pressure Evolution ($p \approx$ const)
Pressure [bar]
Reactor Pressure $p(t)$
Equilibrium $p_{\mathrm{eq}} = 100.6\,\mathrm{bar}$
Initial $p_0 = 100.0\,\mathrm{bar}$
Time [ms]
(d) Isochoric Temperature Relaxation $T(t)$
Temperature [K]
Reactor Temperature $T(t)$
Equilibrium $T_{\mathrm{eq}} = 831.0\,\mathrm{K}$
Initial $T_0 = 1318.7\,\mathrm{K}$
Time [ms]